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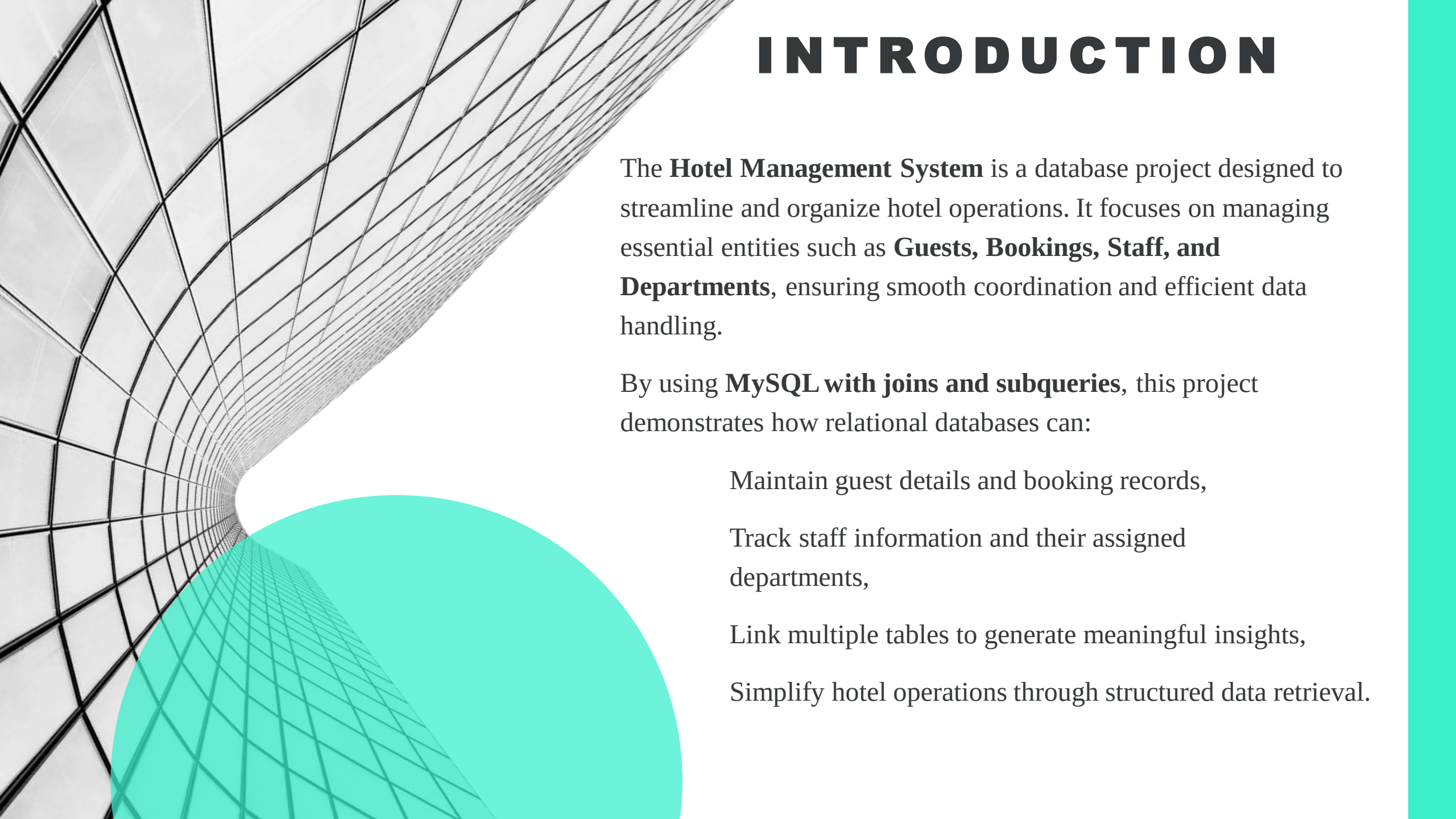
HOTEL MANAGEMENT DATABASE SYSTEM





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INTRODUCTION

The **Hotel Management System** is a database project designed to streamline and organize hotel operations. It focuses on managing essential entities such as **Guests, Bookings, Staff, and Departments**, ensuring smooth coordination and efficient data handling.

By using **MySQL with joins and subqueries**, this project demonstrates how relational databases can:

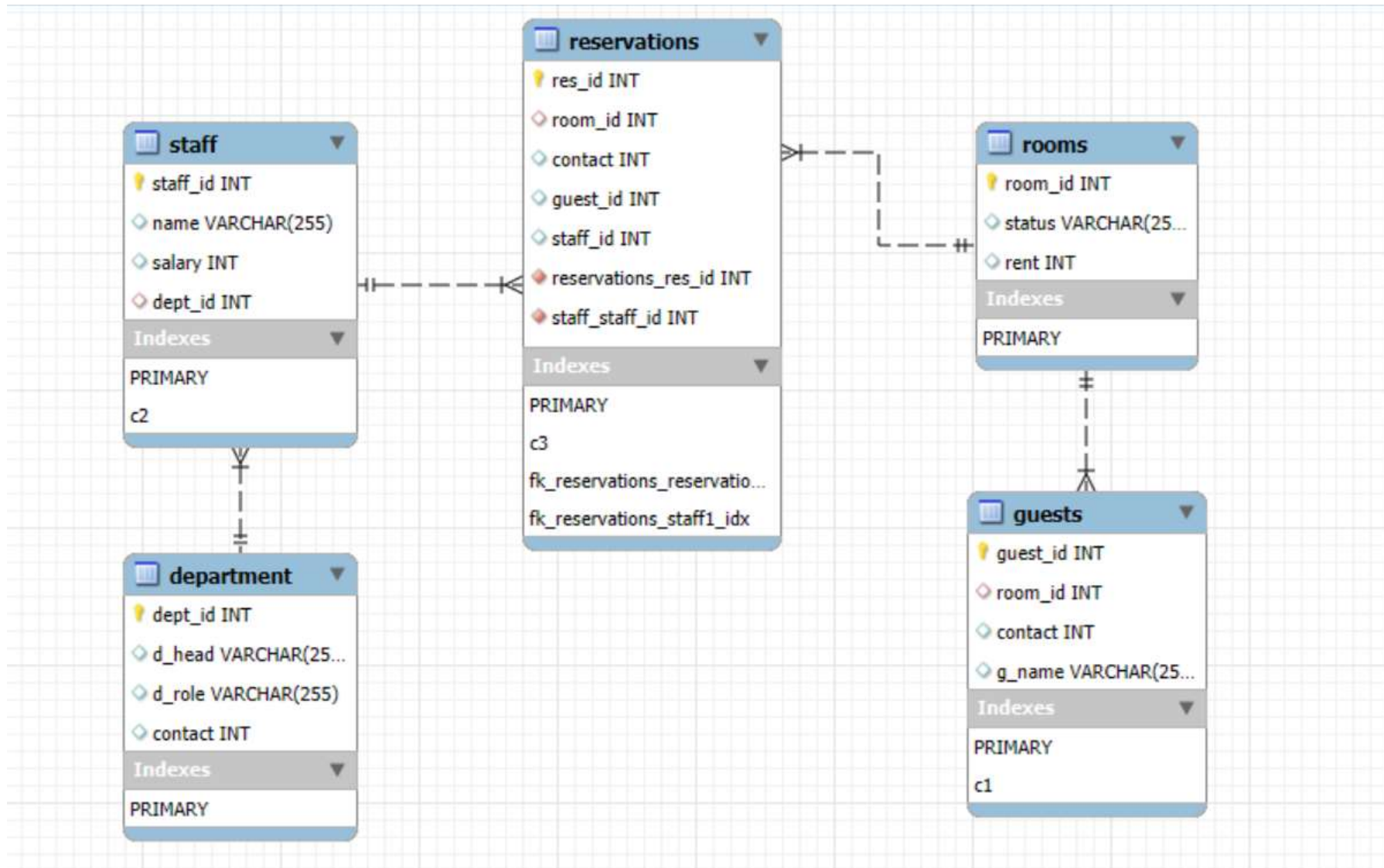
- Maintain guest details and booking records,

- Track staff information and their assigned departments,

- Link multiple tables to generate meaningful insights,

- Simplify hotel operations through structured data retrieval.

ER DIAGRAM



Tables:

Guests: store guest details.

Reservations: Records of reservations made by guests.

Staff: Maintains employee information.

Departments: Categorizes staff into different functional areas

STRUCTURES: ROOMS

```
MySQL localhost:3306 ssl hotel1 SQL > desc rooms;
+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+
| room_id | int           | NO   | PRI | NULL    |       |
| status  | varchar(255)  | YES  |     | NULL    |       |
| rent    | int           | YES  |     | NULL    |       |
| r_type  | varchar(255)  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+
4 rows in set (0.0583 sec)
```

Syntax:

Desc rooms ;

STRUCTURES: STAFF

```
MySQL localhost:3306 ssl hotel1 SQL > desc staff;
```

Field	Type	Null	Key	Default	Extra
staff_id	int	NO	PRI	NULL	
name	varchar(255)	YES		NULL	
salary	int	YES		NULL	
dept_id	int	YES	MUL	NULL	

4 rows in set (0.0015 sec)

Syntax:

`desc staff ;`

STRUCTURES: DEPARTMENT

```
MySQL localhost:3306 ssl hotel1 SQL > desc department;
```

Field	Type	Null	Key	Default	Extra
dept_id	int	NO	PRI	NULL	
d_head	varchar(255)	YES		NULL	
d_role	varchar(255)	YES		NULL	
contact	int	YES		NULL	

```
4 rows in set (0.0026 sec)
```

Syntax:

`desc department ;`

STRUCTURES: GUESTS

```
MySQL localhost:3306 ssl hotel1 SQL > desc guests;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| guest_id   | int           | NO   | PRI | NULL     |       |
| room_id    | int           | YES  | MUL | NULL     |       |
| contact    | int           | YES  |     | NULL     |       |
| g_name     | varchar(255)  | YES  |     | NULL     |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.0020 sec)
```

Syntax:

`desc` guests ;

STRUCTURES: RESERVATIONS

```
MySQL localhost:3306 ssl hotel1 SQL > desc reservations;
+-----+-----+-----+-----+-----+-----+
| Field      | Type  | Null  | Key  | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| res_id     | int   | NO    | PRI  | NULL    |       |
| room_id    | int   | YES   | MUL  | NULL    |       |
| contact    | int   | YES   |      | NULL    |       |
| guest_id   | int   | YES   |      | NULL    |       |
| staff_id   | int   | YES   |      | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.0026 sec)
```

Syntax:

`desc reservations ;`

CONTENTS: Rooms table

```
MySQL localhost:3306 ssl hotel1 SQL > select * from rooms;
```

room_id	status	rent	r_type
1	in use	3000	standard
2	vacant	3000	standard
3	vacant	4500	standard twin
4	in use	4500	executive
5	in use	5500	executive
6	in use	5500	deluxe
7	vacant	6000	executive twin
8	in use	6000	deluxe twin
9	in use	6000	deluxe twin
10	in use	3000	standard
11	in use	3000	standard
12	in use	4500	executive
13	in use	6000	executive twin
14	vacant	6000	executive twin
15	vacant	5500	executive twin
16	in use	6000	deluxe twin
17	vacant	6000	deluxe twin
18	vacant	5500	deluxe
19	vacant	5500	deluxe
20	in use	4500	executive
21	vacant	5500	deluxe twin
22	in use	6000	deluxe
23	in use	5500	executive twin
24	vacant	4000	standard
25	in use	4000	standard twin

```
25 rows in set (0.0011 sec)
```

```
create table rooms (  
  room_id int primary key,  
  status varchar(255),  
  rent int  
  r_type varchar(255)  
);
```

CONTENTS: Guests

```
MySQL localhost:3306 ssl hotel1 SQL > select * from guests;
```

guest_id	room_id	contact	g_name
110	2	1234	Aiden Ross
120	4	5678	Priya Sharma
130	3	3456	Luca Romano
140	5	4353	Naomi Carter
150	6	6893	Hiroshi Tanaka
160	7	56798	Isabella Flores
170	9	45781	Elias Novak
180	10	45678	Sofia Mendes
190	11	1267898	Lila Kapoor
200	12	1267898	Matteo Silva
210	13	1267898	Felix Morgan
220	14	1267898	Amara Joshi
230	15	1267898	Viktor Petrov
250	25	1267898	Jonah Patel
260	22	1267898	Aria Laurent
270	23	1267898	Daniel Kim
340	20	1267898	Serena Clarke

17 rows in set (0.0007 sec)

```
create table guests (  
  guest_id int primary key,  
  room_id int,  
  contact int  
  g_name varchar(255)  
);
```

CONTENTS: Department table

```
MySQL localhost:3306 ssl hotel1 SQL > select * from department;
```

dept_id	d_head	d_role	contact
101	mark	manager	12345
102	jane	security	67890
103	john	front end	23467
104	sam	housekeeping	35790

```
4 rows in set (0.0034 sec)
```

```
MySQL localhost:3306 ssl hotel1 SQL >
```

```
create table department (  
  dept_id int primary key,  
  d_head varchar(255),  
  d_role varchar(255),  
  contact int  
);
```

CONTENTS: Staff table

```
MySQL localhost:3306 ssl hotel1 SQL > select * from staff;
+-----+-----+-----+-----+
| staff_id | name  | salary | dept_id |
+-----+-----+-----+-----+
|      100 | sammy |   4500 |      101 |
|      101 | man   |   3000 |      104 |
|      102 | john  |   5000 |      101 |
|      103 | amy   |   5500 |      101 |
|      104 | ram   |   4500 |      104 |
|      105 | pam   |   3500 |      103 |
|      106 | cam   |   3000 |      102 |
|      107 | tam   |   2500 |      103 |
|      108 | john  |   5000 |      104 |
|      109 | nine  |   4000 |      104 |
|      110 | ten   |   3200 |      104 |
+-----+-----+-----+-----+
11 rows in set (0.0011 sec)
```

```
create table staff (
  staff_id int primary key,
  name varchar(255),
  salary int,
  dept_id int);
```

CONTENTS:Reservations table

MySQL localhost:3306 ssl hotel1 SQL > select * from reservations;

res_id	room_id	contact	guest_id	staff_id	res_name
112	3	5683	110	100	Rohan Iyer
113	2	4358	120	101	Clara Dupont
115	4	3578	130	103	Diego Alvarez
116	2	3452	120	101	Nathaniel Brooks
117	1	1246	120	100	Hana Suzuki
118	3	67523	130	103	Gabriel Torres
119	6	7965	150	103	Stefan Ivanov
120	7	43274	160	102	Evelyn Scott
121	8	342352	190	110	Arjun Malhotra
122	10	534232	190	104	Maya Deshmukh
123	11	534232	200	107	Julian Weber
124	12	534232	210	105	Anika Rao
125	15	534232	220	103	Nisha Verma
126	13	534232	340	102	Tobias Grant
127	10	534232	210	102	Elina Varga

15 rows in set (0.0010 sec)

```
create table reservations (  
  res_id int primary key,  
  room_id int,  
  contact int  
  guest_id int  
  staff_id int  
  res_name varchar(255)  
);
```

JOINS 1: GUESTS AND ROOMS

guest_id	room_id	contact	g_name	room_id	status	rent	r_type
110	2	1234	Aiden Ross	2	vacant	3000	standard
120	4	5678	Priya Sharma	4	in use	4500	executive
130	3	3456	Luca Romano	3	vacant	4500	standard twin
140	5	4353	Naomi Carter	5	in use	5500	executive
150	6	6893	Hiroshi Tanaka	6	in use	5500	deluxe
160	7	56798	Isabella Flores	7	vacant	6000	executive twin
170	9	45781	Elias Novak	9	in use	6000	deluxe twin
180	10	45678	Sofia Mendes	10	in use	3000	standard
190	11	1267898	Lila Kapoor	11	in use	3000	standard
200	12	1267898	Matteo Silva	12	in use	4500	executive
210	13	1267898	Felix Morgan	13	in use	6000	executive twin
220	14	1267898	Amara Joshi	14	vacant	6000	executive twin
230	15	1267898	Viktor Petrov	15	vacant	5500	executive twin
250	25	1267898	Jonah Patel	25	in use	4000	standard twin
260	22	1267898	Aria Laurent	22	in use	6000	deluxe
270	23	1267898	Daniel Kim	23	in use	5500	executive twin
340	20	1267898	Serena Clarke	20	in use	4500	executive

17 rows in set (0.0006 sec)

MySQL localhost:3306 ssl hotel1 SQL >

Syntax:

```
-- joining guests to rooms
```

```
select * from guests g
```

```
inner join rooms r
```

```
on g.room_id = r.room_id;
```

JOINS 2:STAFF AND DEPARTMENT

```
MySQL localhost:3306 ssl hotel1 SQL > select * from staff s right join department d on s.dept_id=d.dept_id;
```

staff_id	name	salary	dept_id	dept_id	d_head	d_role	contact
103	amy	5500	101	101	mark	manager	12345
102	john	5000	101	101	mark	manager	12345
100	sammy	4500	101	101	mark	manager	12345
NULL	NULL	NULL	NULL	102	jane	security	67890
NULL	NULL	NULL	NULL	103	john	front end	23467
101	man	3000	104	104	sam	housekeeping	35790

```
6 rows in set (0.0236 sec)
```

```
MySQL localhost:3306 ssl hotel1 SQL >
```

-- joining staff to department

select * from staff s

right join department d

on s.dept_id = d.dept_id;

JOINS 3: RESERVATIONS AND STAFF

res_id	res_name	name	guest_id	contact
112	Rohan Iyer	sammy	110	3245245
113	Clara Dupont	man	120	75434523
115	Diego Alvarez	amy	130	25363232
116	Nathaniel Brooks	man	120	75434523
117	Hana Suzuki	sammy	120	3245245
118	Gabriel Torres	amy	130	25363232
119	Stefan Ivanov	amy	150	25363232
120	Evelyn Scott	john	160	679653
121	Arjun Malhotra	ten	190	78534242
122	Maya Deshmukh	ram	190	768564435
123	Julian Weber	tam	200	45645765
124	Anika Rao	pam	210	43224342
125	Nisha Verma	amy	220	25363232
126	Tobias Grant	john	340	679653
127	Elina Varga	john	210	679653

15 rows in set (0.0012 sec)

MySQL localhost:3306 ssl hotel1 SQL >

-- (who handled the booking ?)

```
select r.res_id, g.guest_id, s.name
from reservations r
inner join guests g
on r.guest_id = g.guest_id
inner join staff s
on r.staff_id = s.staff_id;
```

SUBQUERIES 1:

Guest	room_id	r_type	rent
Isabella Flores	7	executive twin	6000
Lila Kapoor	8	deluxe twin	6000
Serena Clarke	13	executive twin	6000

3 rows in set (0.0015 sec)

MySQL localhost:3306 ssl hotel1 SQL >

--Who has the most expensive room?

```
SELECT g.g_name AS Guest, r.room_id, rm.r_type, rm.rent
FROM Guests g
JOIN Reservations r
ON g.guest_id = r.guest_id
JOIN Rooms rm
ON r.room_id = rm.room_id
WHERE rm.rent = ( SELECT MAX(rent) FROM Rooms);
```

SUBQUERIES 2:

r_type	total_revenue
standard	18000
executive twin	17500
standard twin	9000
executive	9000
deluxe twin	6000
deluxe	5500

6 rows in set (0.0007 sec)

MySQL localhost:3306 ssl hotel1 SQL >

-- revenue per room type

```
SELECT rm.r_type, SUM(res.rent) AS total_revenue
FROM Reservations res
JOIN Rooms rm
ON res.room_id = rm.room_id
GROUP BY rm.r_type
ORDER BY total_revenue DESC;
```

SUBQUERIES 3:

r_type	available_rooms
deluxe	18,19
deluxe twin	17,21
executive twin	7,14,15
standard	2,24
standard twin	3

5 rows in set (0.0017 sec)

-- Available rooms according to room type

```
SELECT r_type, GROUP_CONCAT(room_id) AS  
available_rooms
```

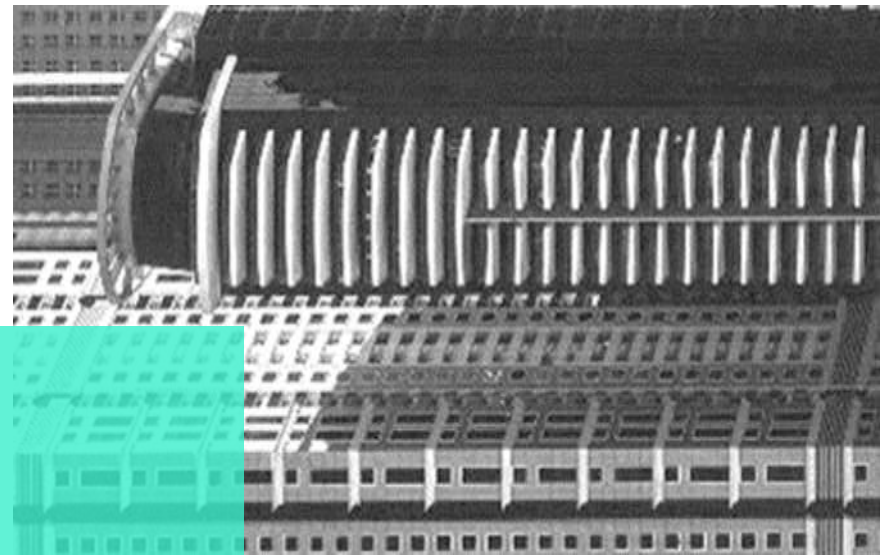
```
FROM Rooms
```

```
WHERE status = 'vacant'
```

```
GROUP BY r_type;
```

SUMMARY

The Hotel Management System project shows how MySQL databases can streamline hotel operations by organizing guests, bookings, staff, reservations and department data efficiently. Through the use of joins and subqueries, the system demonstrates the power of relational databases in providing quick, accurate, and connected information essential for smooth hotel management.





THANK YOU

Shivangi bhagurkar